`GeoModels` Tutorial: simulation, estimation and prediction of spatial data with heavy tails using Tukey-h random fields

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Introduction

In this tutorial we show how to analyze spatial data with heavy tails using Tukey-h random fields [Xu and Genton, 2017] with the `R` package `GeoModels` [Bevilacqua et al., 2026]. The Tukey-h distribution is a flexible parametric model, which is able to accommodate flexible tail behaviour and, in particular, heavier tails than the ones induced by Gaussian random fields.

We first load the `R` libraries needed in this tutorial and set the name of the model in the `GeoModels` package.

```
rm(list=ls())
require(GeoModels)
require(fields)
model="Tukeyh" # model name in the GeoModels package
set.seed(818)
```

Simulation of Tukey-h random fields

The definition of a Tukey-h random field starts by considering a 'parent' Gaussian random field $G = \{G(\boldsymbol{s}), \boldsymbol{s} \in S\}$, where $\boldsymbol{s}$ represents a location in the domain S. In this tutorial we consider the spatial case, i.e., $S \subseteq \mathbb{R}^2$. However, the package `GeoModels` allows to work also with spatio-temporal data or data defined on a sphere of arbitrary radius. The Gaussian field G is assumed weakly stationary with zero mean, unit variance and correlation function $\rho(\boldsymbol{u}) = \mathrm{cor}\{G(\boldsymbol{s}+\boldsymbol{u}), G(\boldsymbol{s})\}$.

Let $T_h^* = \{T_h^*(\boldsymbol{s}), \boldsymbol{s} \in S\}$ be a random field with standard Tukey-h marginal distribution defined through a monotonic transformation $\tau_h(x) = x \exp(hx^2/2)$, $x \in \mathbb{R}$, of a standard Gaussian RF G as:

$$T_h^*(\boldsymbol{s}) := \tau_h\{G(\boldsymbol{s})\} = G(\boldsymbol{s}) \exp\left\{\frac{hG(\boldsymbol{s})^2}{2}\right\}. \tag{1}$$

The inverse transformation $\tau_h^{-1}(t)$ can be expressed in terms of the Lambert function, i.e., $\tau_h^{-1}(t) = \mathrm{sign}(t)\{W(ht^2)/h\}^{1/2}$ for $h > 0$, where $W(\cdot)$ is the Lambert-W function [Goerg, 2015]; for $h = 0$ the inverse is t.

This kind of random field has marginal symmetric distributions and the parameter $h \in [0, 1/2)$ governs its tail behavior, with a larger value of h indicating a heavier tail.

Specifically, the marginal distribution has (asymptotically) a Pareto-heavy tailed distribution with tail index equal to $1/h$ [Morgenthaler and Tukey, 2000]. If $h = 0$ the Gaussian random field G is obtained as a special limit case.

In addition, the Tukey-h RF has marginal distribution given by Goerg [2015]:

$$f_{T_h^*}(t) = \frac{\tau_h^{-1}(t)}{t\{1 + W(ht^2)\}}\phi\{\tau_h^{-1}(t); 0, 1\}, \tag{2}$$

where the expression at $t = 0$ is interpreted by continuity. Its moments are $\mathrm{E}\{T_h^*(\boldsymbol{s})\} = 0$, $\mathrm{var}\{T_h^*(\boldsymbol{s})\} = (1 - 2h)^{-3/2}$ and the correlation function is:

$$\rho_{T_h^*}(\boldsymbol{u}) = \frac{\rho(\boldsymbol{u})(1 - 2h)^{3/2}}{[(1 - h)^2 - h^2\rho^2(\boldsymbol{u})]^{3/2}}. \tag{3}$$

Then, we define the location-scale transformation process $T_h = \{T_h(\boldsymbol{s}), \boldsymbol{s} \in S\}$ as:

$$T_h(\boldsymbol{s}) := \mu(\boldsymbol{s}) + \sigma T_h^*(\boldsymbol{s}) \tag{4}$$

with $\mathrm{E}\{T_h(\boldsymbol{s})\} = \mu(\boldsymbol{s})$ and $\mathrm{var}\{T_h(\boldsymbol{s})\} = \sigma^2(1 - 2h)^{-3/2}$. A spatial regression model can be specified by assuming that $\mu(\boldsymbol{s}) = X(\boldsymbol{s})^T\boldsymbol{\beta}$ where $X(\boldsymbol{s})$ is a k-dimensional vector of covariates and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_k)^T$ is a k-dimensional vector of unknown parameters. In this tutorial we assume $k = 2$.

To obtain a simulation from T_h we need to specify regression mean, tail and variance parameters, i.e., β_1, β_2, h, and σ^2, respectively. Additionally, we need a parametric correlation $\rho(\boldsymbol{u})$ for the 'parent' Gaussian RF. We first set the spatial coordinates. In this example we consider 1500 locations uniformly distributed in the unit square:

```
N=1500
coords=cbind(runif(N),runif(N))
plot(coords,pch=20,xlab="",ylab="")
```

For the correlation function $\rho(\boldsymbol{u})$ of the 'parent' Gaussian random field G we assume an isotropic Matérn model [Matérn, 1986]:

$$\rho_{\alpha,\gamma}(\boldsymbol{u}) = \frac{2^{1-\gamma}}{\Gamma(\gamma)}\left(\frac{\|\boldsymbol{u}\|}{\alpha}\right)^{\gamma} K_\gamma\left(\frac{\|\boldsymbol{u}\|}{\alpha}\right), \quad \|\boldsymbol{u}\| \geq 0, \tag{5}$$

where K_γ is a modified Bessel function of the second kind of order γ, $\gamma > 0$ is the smoothness parameter and $\alpha > 0$ the spatial scale parameter. Then we set the parameters associated with this correlation model:

```
corrmodel="Matern" ## correlation model
scale=0.2/3 ## scale parameter
smooth=0.5 ## smooth parameter
nugget=0 ## nugget parameter
```

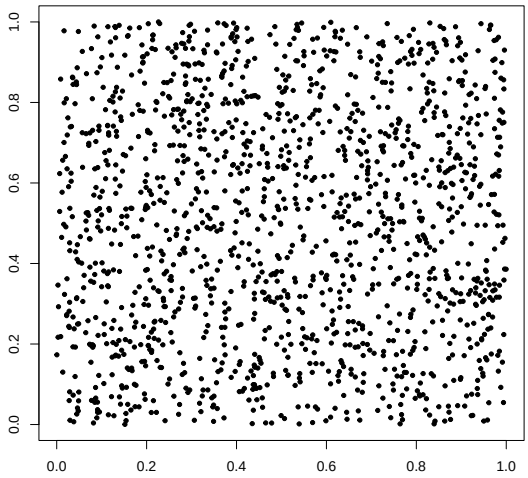


Figure 1: Spatial location sites used in the tutorial.

and we set the tail and variance parameters of the Tukey-h random field:

```
tail=0.1; # tail parameter
sill=1; # variance-scale parameter sigma^2
```

Finally we set the mean regression parameters and the regression matrix:

```
mean=0.5; mean1=-1 # regression parameters
a0=rep(1,N); a1=runif(N,-1,1)
X=cbind(a0,a1) ## regression matrix
```

We are now ready to simulate a realization of the Tukey-h random field T_h using the function `GeoSim`. Simulation is performed exploiting the stochastic representation (1), where the underlying Gaussian field is generated with Cholesky decomposition:

```
param=list(nugget=nugget,mean=mean,mean1=mean1,
    scale=scale,smooth=smooth,sill=sill,tail=tail)
data <- GeoSim(coordx=coords,corrmodel=corrmodel,
    param=param,model=model,X=X)$data
```

Estimation of Tukey-h random fields

Estimation of regression, tail and correlation parameters of the Tukey-h RF T_h can be performed using conditional pairwise likelihood estimation. Let $f_{T_h^*;ij}(t_i, t_j)$ be the density of the bivariate random vector $(T_h^*(\boldsymbol{s}_i), T_h^*(\boldsymbol{s}_j))^T$ given by:

$$f_{T_h^*;ij}(t_i, t_j) = \frac{\tau_h^{-1}(t_i)\tau_h^{-1}(t_j)}{t_i t_j\{1 + W(ht_i^2)\}\{1 + W(ht_j^2)\}}\phi_2\big(\boldsymbol{\tau}_h^{-1}(\boldsymbol{t}); \boldsymbol{0}, R_2\big) \tag{6}$$

with $R_2 = [\rho(\boldsymbol{s}_i - \boldsymbol{s}_j)]_{i,j=1}^2$, the correlation matrix associated with the underlying Gaussian correlation function, and τ_h^{-1} applied pointwise to $\boldsymbol{t} = (t_i, t_j)^T$.

Given a partial realization $(t(\boldsymbol{s}_1), \ldots, t(\boldsymbol{s}_N))^T$ of the Tukey-h random process T_h defined in (4), the density of the bivariate vector $(T_h(\boldsymbol{s}_i), T_h(\boldsymbol{s}_j))^T$ can be obtained from (6) as:

$$f_{T_h;ij}\{t(\boldsymbol{s}_i), t(\boldsymbol{s}_j)\} = \frac{1}{\sigma^2} f_{T_h^*;ij}\left(\frac{t(\boldsymbol{s}_i) - \mu(\boldsymbol{s}_i)}{\sigma}, \frac{t(\boldsymbol{s}_j) - \mu(\boldsymbol{s}_j)}{\sigma}\right). \tag{7}$$

Then, the conditional pairwise likelihood function is defined as:

$$cpl(\boldsymbol{\theta}) = \sum_{i=1}^{N}\sum_{\substack{j=1 \\ j\neq i}}^{N} \log\left\{\frac{f_{T_h;ij}\{t(\boldsymbol{s}_i), t(\boldsymbol{s}_j)\}}{f_{T_h}\{t(\boldsymbol{s}_j)\}}\right\} w_{ij}. \tag{8}$$

Here w_{ij} are non-negative weights, not depending on $\boldsymbol{\theta}$. An efficient way to specify the (non-symmetric) weights from a computational viewpoint is based on neighborhoods:

$$w_{ij}(k) = \begin{cases} 1 & \boldsymbol{s}_i \in N_k(\boldsymbol{s}_j), \\ 0 & \text{otherwise.} \end{cases} \tag{9}$$

Here $N_k(\boldsymbol{s}_l)$ is the set of the k nearest neighbors of the point $\boldsymbol{s}_l$ and in this case $\boldsymbol{\theta} = (\beta_1, \beta_2, h, \sigma^2, \alpha, \gamma)^T$. The conditional pairwise likelihood estimator $\widehat{\boldsymbol{\theta}}_{cpl}$ is obtained by maximizing (8) with respect to $\boldsymbol{\theta}$. In the `GeoModels` package we can choose the fixed and estimated parameters.

Conditional pairwise likelihood estimation can be performed using `GeoFit` or `GeoFit2`. In particular, `GeoFit2` first attempts an independence marginal fit to initialize the marginal parameters and then performs conditional pairwise likelihood estimation using these starting values. If the preliminary step is unavailable or fails, the supplied starting values are retained.

We maximize (8) with `nlminb`, which allows box-constrained optimization using PORT routines.

```
optimizer="nlminb"
fixed1<-list(nugget=nugget,smooth=smooth)
start1<-list(mean=mean,mean1=mean1,
            scale=scale,sill=sill,tail=tail)
I=Inf
lower1<-list(mean=-I,mean1=-I,
            scale=1e-6,sill=1e-6,tail=0)
upper1<-list(mean=I,mean1=I,
            scale=I,sill=I,tail=0.5-1e-6)
fit2 <- GeoFit2(data=data,coordx=coords,
   corrmodel=corrmodel,optimizer=optimizer,
   lower=lower1,upper=upper1,type="Pairwise",
   likelihood="Conditional",neighb=4,X=X,
   start=start1,fixed=fixed1,model=model)
```

Note that the option `neighb=4` sets the neighborhood order in the weight function (9), i.e., $k = 4$.

The object `fit2` includes information about the conditional pairwise likelihood fit:

```
fit2
# Historical output from the original tutorial:
# Maximum Composite-Likelihood Fitting of Tukeyh RF
# Setting: Conditional Composite-Likelihood
# Covariance model: Matern; Optimizer: nlminb
# Maximum log-Composite-Likelihood value: -6848.66
# mean mean1 scale sill tail
# 0.43734 -1.00500 0.06962 1.02365 0.11640
```

Checking model assumptions

Given the estimates of the mean regression and sill parameters, the estimated residuals

$$\widehat{T_h^*}(\boldsymbol{s}_i) = \frac{t(\boldsymbol{s}_i) - X(\boldsymbol{s}_i)^T \widehat{\boldsymbol{\beta}}}{\sqrt{\widehat{\sigma^2}}}, \quad i = 1, \ldots, N,$$

can be viewed as a realization of the process T_h^*. The residuals can be computed using the `GeoResiduals` function:

```
res=GeoResiduals(fit2)
```

The marginal distribution assumption on residuals can be graphically checked with a QQ-plot using the `GeoQQ` function (see Figure 2, left part):

```
GeoQQ(res)
GeoQQ(res,type="D",ylim=c(0,0.5),breaks=20)
```

The covariance model assumption can be checked comparing empirical and estimated semi-variograms using `GeoVariogram` and `GeoCovariogram` (see Figure 2, right part):

```
vario <- GeoVariogram(data=res$data,
                              coordx=coords,maxdist=0.4)
GeoCovariogram(res,show.vario=TRUE,
                        vario=vario,pch=20)
```

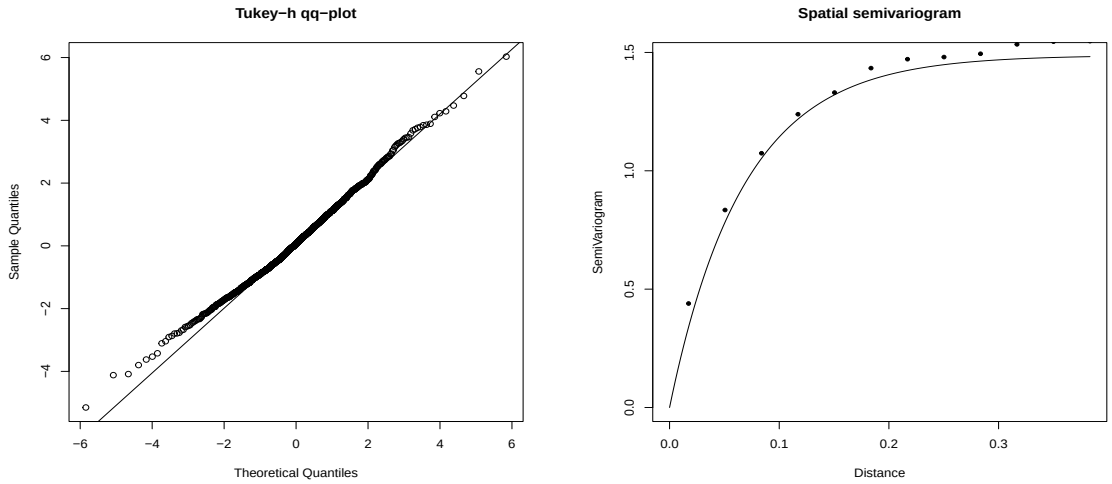


Figure 2: From left to right: QQ-plot for residuals using the Tukey-h distribution with estimated tail $\widehat{h} \simeq 0.1164$ and empirical versus estimated semi-variogram for the residuals (original tutorial).

The semi-variogram is computed using the correlation function (3).

Prediction of Tukey-h random fields

For a given location $\boldsymbol{s}_0$ with associated covariates $X(\boldsymbol{s}_0)$, the optimal linear predictor of a Tukey-h random field is given by:

$$\widehat{T_h}(\boldsymbol{s}_0) = X(\boldsymbol{s}_0)^T\boldsymbol{\beta} + \sum_{i=1}^{N} \lambda_i \left[t(\boldsymbol{s}_i) - X(\boldsymbol{s}_i)^T\boldsymbol{\beta}\right] \tag{10}$$

where the vector $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_N)^T$ is given by $\boldsymbol{\lambda} = R_h^{-1}\boldsymbol{c}_h$ and

- $\boldsymbol{c}_h = (\text{cor}\{T_h(\boldsymbol{s}_0), T_h(\boldsymbol{s}_1)\}, \ldots, \text{cor}\{T_h(\boldsymbol{s}_0), T_h(\boldsymbol{s}_N)\})^T$.
- $R_h = [\text{cor}\{T_h(\boldsymbol{s}_i), T_h(\boldsymbol{s}_j)\}]_{i,j=1}^N$ is the correlation matrix.

Moreover the associated mean squared error (MSE) is given by:

$$MSE\{\widehat{T_h}(\boldsymbol{s}_0)\} = \sigma^2(1-2h)^{-3/2}\left(1 - \boldsymbol{c}_h^T R_h^{-1}\boldsymbol{c}_h\right). \tag{11}$$

This is the best linear predictor based on the mean and correlation function (3); in the non-Gaussian case it need not coincide with the conditional mean. If parameters are unknown, (10) and (11) are computed replacing them with conditional pairwise likelihood estimates. In particular R_h and $\boldsymbol{c}_h$ depend on the estimated Matérn correlation function and the estimated tail parameter h.

Kriging and associated MSE can be obtained using `GeoKrig`. We first need to specify the spatial locations to predict and, in this example, consider a regular grid:

```
xx=seq(0,1,0.012)
loc_to_pred=as.matrix(expand.grid(xx,xx))
Nloc=nrow(loc_to_pred)
Xloc=cbind(rep(1,Nloc),runif(Nloc))
```

Then the optimal linear prediction (10), using fitted parameters, can be obtained with (computation can be time consuming):

```
pr=GeoKrig(fit2,loc=loc_to_pred,
                X=X,Xloc=Xloc,mse=TRUE)
```

Finally, a kriging map with associated MSE (Figure 3) can be obtained:

```
quilt.plot(coords[,1],coords[,2],data,main="Data")
map=matrix(pr$pred,ncol=length(xx))
image.plot(xx,xx,map,xlab="",ylab="",main="SimpleKriging")
map_mse=matrix(pr$mse,ncol=length(xx))
image.plot(xx,xx,map_mse,xlab="",ylab="",main="MSE")
```

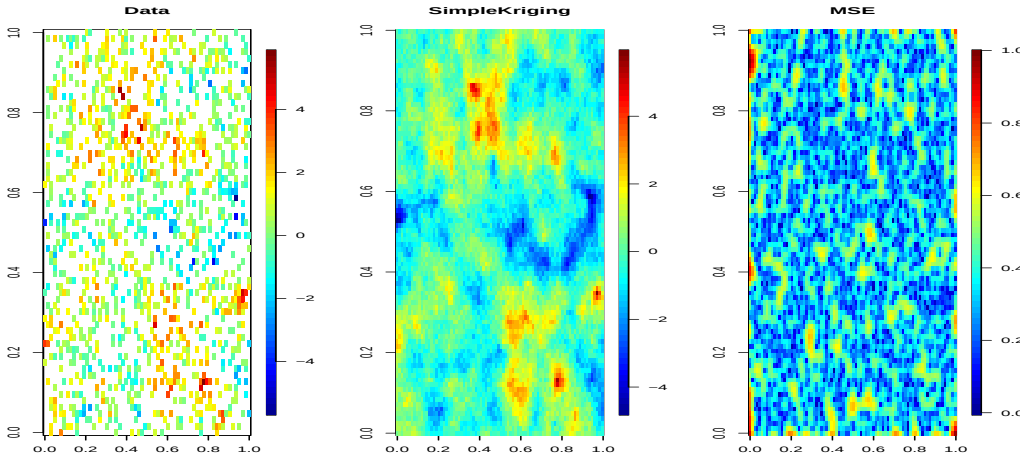


Figure 3: From left to right: observed spatial data, associated kriging map and mean square error map (original tutorial).

References

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